

Computer Network - Detailed Five Mark Answers

1. What is a computer network?

A computer network is a collection of two or more computers and other devices connected together to share data, information, and resources. Networks allow users to communicate easily and share devices such as printers, scanners, and internet connections. Computer networks are widely used in schools, offices, banks, hospitals, and homes. Networking improves communication, saves time, and increases efficiency in work.

2. What are the advantages of a computer network?

A computer network has many advantages. It allows users to share files and information quickly. Hardware devices such as printers and scanners can also be shared among many users, which reduces cost. Communication becomes easier through emails, chats, and video calls. Networks also help in sharing internet connections and improve teamwork. Data can be stored centrally, making management and backup easier.

3. Name the devices used to set up a computer network.

The devices used to set up a computer network include router, switch, hub, modem, and network cables. A router connects different networks and provides internet access. A switch connects multiple devices within the same network. A hub also connects devices but is less efficient than a switch. A modem connects the network to the internet service provider. Network Interface Cards (NICs) and cables are also used for communication between devices.

4. What is the significance of a server in a computer network?

A server is an important and powerful computer in a network that provides services and resources to other computers called clients. It stores data, manages files, controls access to resources, and handles network security. A server can also provide internet access, printing services, and application services. It helps in central management of information and improves the efficiency and security of the network.

5. Distinguish between LAN, MAN and WAN.

LAN	MAN	WAN
Covers a small area	Covers a city or town	Covers very large areas
Used in schools, homes, and offices	Used within a city	Used between countries and continents
High speed and low cost	Moderate speed and cost	Lower speed and expensive
Example: School computer lab	Example: Cable TV network	Example: Internet

6. What is real-time communication?

Real-time communication is the exchange of information instantly without delay between two or more users. In this type of communication, the sender and receiver communicate at the same time. Examples of real-time communication include video conferencing, online chatting, voice calls, and live streaming. It is useful in business meetings, online classes, and customer support services.

7. What is a client in a computer network?

A client is a computer or device in a network that receives services and resources from a server. Clients use shared files, applications, internet connections, and printers provided by the server. In a client-server network, clients send requests to the server and the server responds to those requests. Desktop computers, laptops, and smartphones connected to a network can act as clients.

8. Distinguish between Internet, Intranet and Extranet.

Internet	Intranet	Extranet
Public network	Private network	Limited private network
Accessible to everyone	Accessible only within an organization	Accessible to selected outsiders
Worldwide communication	Used for internal communication	Used for business communication
Example: Google	Example: School office network	Example: Online supplier network

9. What are the advantages of e-commerce?

E-commerce has many advantages for both customers and businesses. It allows people to buy and sell products online from anywhere at any time. It saves time and travelling expenses. Customers can compare prices and choose from a wide variety of products. Online payment methods make transactions easy and fast. Businesses can reach customers worldwide and provide services 24 hours a day.

10. Answers related to network hardware.

- (a) Each computer connected in a network is called a node or workstation.
- (b) The powerful computer that provides services to other computers is called a server.
- (c) A switch or hub is used to connect computers, printers, and other devices together in a network.
- (d) Networking hardware such as routers, switches, cables, and NICs are used to connect different devices in a computer network.